02

THE WORM BRAIN

400-million-year-old survival circuit, rebuilt in integer math.

BODHI · A Conscious Digital Brain

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BODHI · REPORT 02

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BODHI — A Conscious Digital Brain
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Patents: IP Australia 2026901656 (image) / 2026901657 (audio)

SK

FOUNDATIONS

The worm brain

At the base of BODHI sits a faithful simulation of the C. elegans connectome — 26 neurons, 39 synapses, int8-range weights (clamped to −127..+127, hand-tuned in −60..+80 today). It is the oldest known neural circuit: 400 million years of evolutionary debugging, compressed into less than a thousand lines of Python.

Why a worm

Because a worm brain is the only biological neural network that has been mapped to the synapse. Every connection has been traced. Every neuron has a name. And every behavior — sensing food, escaping heat, reversing, pausing — can be reproduced in code. If you cannot simulate a worm, you cannot simulate anything bigger.

The numbers

26
neurons (8 sensory + 12 inter + 6 motor)

39
directed synapses, int8-range weights

7/7
reflex tests pass in selftest

<1 ms
per tick on CPU

Architecture

- Sensory layer (8). Chemosensors, thermosensors, mechanosensors — map external stimuli to integer activation currents (0–255).
- Interneuron layer (12). Command and coordination neurons — AVA, AVB, AVD, AVE, PVC and friends.
- Motor layer (6). Forward, backward, turn, pause. Pure actuation.
- Weights are integer. No float math. Every multiplication is a shift-and-add. Deterministic on every CPU.
- Updates are event-driven. No global clock. A synapse fires when upstream activation crosses threshold.

How BODHI uses it

The worm circuit is not a toy. It is the survival kernel that sits under everything else. Every drive (hunger, curiosity, safety) is implemented as an upstream input to the worm's AVA/AVB command neurons. Every motor plan — even one generated by Broca to move a BODHI character in the Phase 2 simulation playground — is gated by the worm's

forward/backward/pause logic. This means the full BODHI brain is always obeying a 26-neuron survival circuit that never fails.

> Design rule. If the worm circuit tells you to back off, you back off. The worm has veto power over every other module. This is how we guarantee graceful failure.

Sample trace

```
tick=000 AVA=+12 AVB=+02 PVC=-04 motor=forward
tick=001 AVA=+41 AVB=+05 PVC=+19 motor=forward (accelerate)
tick=002 AVA=+72 AVB=-11 PVC=+38 motor=forward
tick=003 AVA=+99 AVB=-44 PVC=+57 motor=pause (threshold)
tick=004 AVA=-23 AVB=-61 PVC=+48 motor=reverse (flight)
tick=005 AVA=-47 AVB=-82 PVC=+12 motor=reverse
```

Tests

Reflex	Input	Expected	Result
Food gradient	chem+=50	forward, accelerate	PASS
Heat shock	therm+=80	reverse within 3 ticks	PASS
Nose touch	mech+=60	pause then reverse	PASS
Starvation	chem=0 for 50t	increase curiosity drive	PASS
Harsh light	photo+=70	omega-turn	PASS
Gentle touch	mech+=15	pause only	PASS
Dual gradient	chem+50 / therm+40	resolve via AVB	PASS